FORM TP 2024196

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MAY/JUNE 2024

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

CHEMISTRY

UNIT 2 – Paper 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of THREE questions. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. Do NOT write in the margins.
4. Where appropriate, ALL WORKING MUST BE SHOWN in this booklet.
5. A data booklet is provided.
6. You may use a silent, non-programmable calculator to answer questions.
7. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
8. **If you use the extra page(s), you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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Answer ALL questions.

MODULE 1: THE CHEMISTRY OF CARBON COMPOUNDS

1. A hydrocarbon compound whose molecular formula is C_8H_{18} was determined to have a given number of isomers.

(a) (i) Define the term 'structural isomerism'.

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[2 marks]

(ii) List THREE types of structural isomerism.

.....

.....

[3 marks]

(iii) Define ONE of the types of structural isomerism listed in (a) (ii).

.....

.....

[2 marks]

(b) (i) Draw TWO isomers that have ONE methyl substituent.

[2 marks]

(ii) Draw THREE isomers that contain ONLY secondary and tertiary carbon atoms and at least TWO methyl substituents.

[3 marks]

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(c) The scheme in Figure 1 shows a sequence of reactions that indicate the outcome of reacting an isomer of Hydrocarbon A. Carefully examine the scheme and answer the questions that follow.

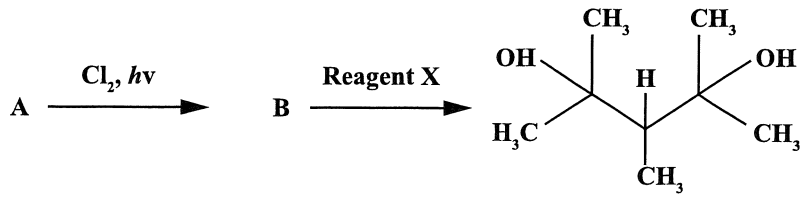

Figure 1. Scheme showing reaction sequence of Hydrocarbon A

(i) Give the structural formula of Compound A, Compound B and Reagent X.

Compound A

[2 marks]

Compound B

[2 marks]

Reagent X .....

[1 mark]

(ii) List the names of the steps involved in going from A to B in Figure 1.

.....

.....

.....

[3 marks]

GO ON TO THE NEXT PAGE

(iii) Using curved arrows, give the mechanism for the first step listed in (c) (ii) **on page 6**.

[2 marks]

(iv) Give the structural formula of a product that can be formed alternatively to Compound B in (c) (i).

[1 mark]

(v) Given that ethanol has a pKa of 16, suggest whether the pKa of Compound C is higher or lower than that of ethanol. Give a reason for your suggestion.

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[3 marks]

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(vi) If the reaction of going from A to B was performed in a closed vessel, what would be the expected colour change?

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[2 marks]

(vii) In an experiment, two drops of universal indicator were added to 1 cm^3 of ethanol in Test tube 1 and 1 cm^3 of Compound C in Test tube 2.

Suggest the colours expected in Test tubes 1 and 2.

Test tube 1 .....

Test tube 2 .....

[2 marks]

Total 30 marks

MODULE 2: ANALYTICAL METHODS AND SEPARATION TECHNIQUES

2. A liquid sample of an unknown organic compound was submitted to a lab facility for structural analysis. The sample was irradiated with radiation from various parts of the electromagnetic spectrum, labelled A–D, as seen in Figure 2.

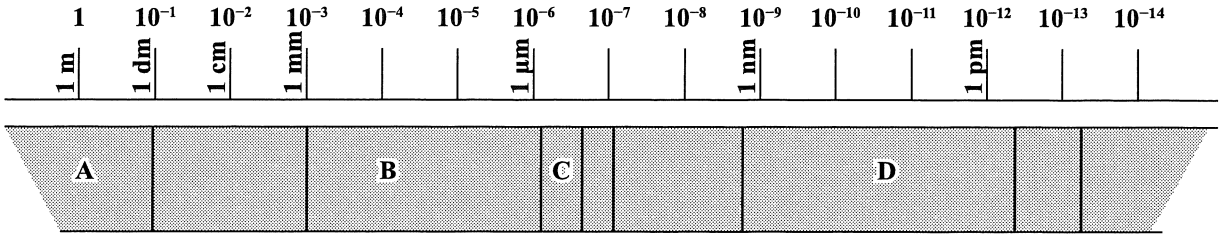

Figure 2. Electromagnetic spectrum of unknown organic compound

(a) Give the name of EACH radiation type.

(i) A .....

(ii) B .....

(iii) C .....

(iv) D .....

[4 marks]

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(b) Spectra obtained from the excitation of the sample showed an absorption signal at 9.4×10^{14} Hz.

(i) Calculate the wavelength, λ, in metres (m), for this radiation.

[3 marks]

(ii) With reference to Figure 2, in what radiation band does the calculated wavelength in (b) (i) belong?

.....

[1 mark]

(iii) A lamp with filament made from the sample emits light of the same wavelength, λ, obtained in (b) (i). Calculate the energy of a single emitted photon.

[3 marks]

(c) Acetylsalicylic acid is a well-known drug that is commonly known as aspirin. To determine the concentration of acetylsalicylic acid in a sample of unknown concentration, a student ground ten aspirin tablets into a fine powder and dissolved it in water to make a stock solution of 250 cm^3 for titration. Thereafter, 25.0 cm^3 of the solution was titrated with an excess of standard NaOH. The equation and table below show the reaction between aspirin and standard NaOH along with two possible titration options.

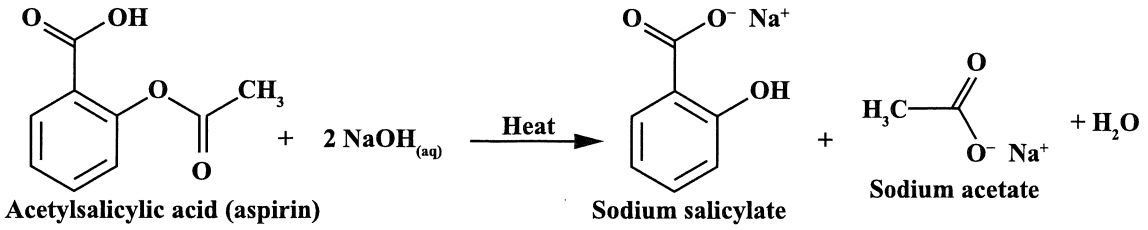

Figure 3. Equation for Option 1

TABLE 1: TITRATION OF ACETYLSALICYLIC ACID

Option	Equation	Titration Method
1	$CH_3COOC_6H_4COOH + 2NaOH \rightarrow CH_3COONa + HOC_6H_4COONa + H_2O$	Titration of excess 0.100 M NaOH with 0.100 mol dm^{-3} acid
2	$CH_3COOC_6H_4COOH + NaOH \rightarrow CH_3COOC_6H_4COONa + H_2O$	Titration of acetylsalicylic acid with 0.100 mol dm^{-3} NaOH

(i) State which acid would be appropriate to use in Option 1.

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[1 mark]

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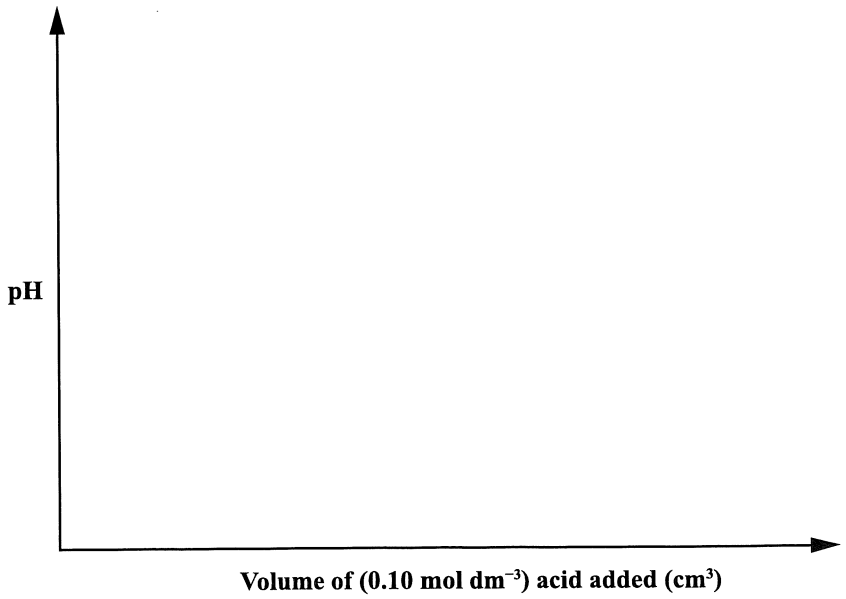
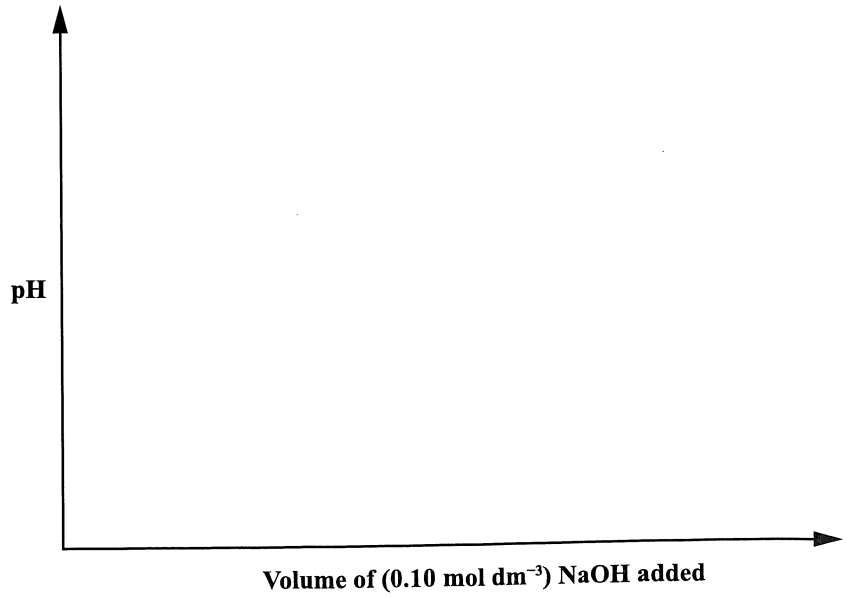
(ii) In the space below, draw the titration curve that would be obtained for Option 1.



[3 marks]

(iii) In the space below, draw the titration curve that would be obtained for Option 2.



[3 marks]

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(iv) Using the information in Table 2 below, select the indicator(s) that **cannot** be used for the titrations.

TABLE 2: TABLE OF COMMONLY USED INDICATORS AND THEIR pH RANGES

Methyl orange	Phenolphthalein	Bromothymol blue	Methyl violet
3.1–4.4	8.2–10.0	6.0–7.6	0.0–1.0

Option 1 .....

[1 mark]

Reason for selection

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[1 mark]

Option 2 .....

[1 mark]

Reason for selection

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[1 mark]

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(v) The titration results obtained from Option 2 are listed below. Complete the table by filling in the spaces.

	First Accurate Titration	**Second Accurate Titration**	**Third Accurate Titration**
Final reading (cm^3)	30.80	29.00	30.00
Initial reading (cm^3)	11.30	9.70	10.80
Titre (cm^3)			
Mean titre (cm^3)			

[2 marks]

(vi) Given that the stock solution of 250 cm^3 contained ten tablets, use the information in (c) (v) to calculate the number of milligrams of aspirin contained in a titrated volume of 25.0 cm^3 of aspirin (MW = 180.06 g/mol).

[4 marks]

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(vii) The ingredient table below lists the milligram quantities of aspirin in a single tablet.

TABLE 3: MILLIGRAM QUANTITIES OF ACTIVE INGREDIENT IN A SINGLE TABLE OF ASPIRIN

Active Ingredient	Acceptable Range in One Tablet	Unit
Acetylsalicylic acid (aspirin)	285–315	mg
For standard strength aspirin tablets		

Based on the findings in (c) (vi), state whether the information in Table 3 is accurate and give a reason for your response.

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[2 marks]

Total 30 marks

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MODULE 3: INDUSTRY AND THE ENVIRONMENT

3. The industrial synthesis of ammonia, known as the Haber–Bosch process, is depicted in the Figure 4 below.

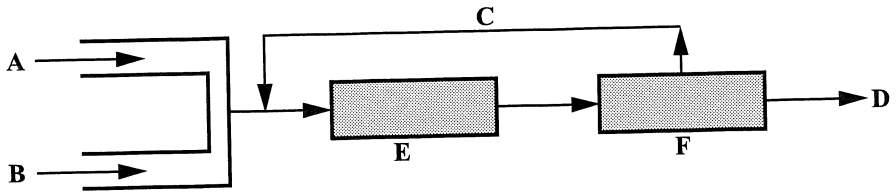

Figure 4. Haber–Bosch synthesis of ammonia

(a) Identify the species represented by the labels A, B, C and D in Figure 4.

A .....

B .....

C .....

D .....

[5 marks]

(b) Use an equation with state symbols to indicate the process that occurs in Vessel E in Figure 4.

E .....

[2 marks]

(c) Use chemical formula and state symbols to indicate the process that occurs in Vessel F in Figure 4.

F .....

[2 marks]

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(d) The reaction to form the ammonia product is favoured towards the formation of the product under certain reaction conditions.

(i) State Le Chatelier's principle and use it to explain how increases in the pressure of the reactants support the formation of the ammonia product in the Haber–Bosch process.

Le Chatelier's principle

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[2 marks]

Explanation

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[4 marks]

(ii) Identify the unique feature of the starting reactants that is mostly responsible for the exothermic nature of the reaction.

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[1 mark]

(iii) State the reaction conditions that are specific to the Haber–Bosch process.

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[3 marks]

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(iv) Indicate how EACH component of the conditions stated in (d) (iii) increases the rate of collisions of reactant molecules.

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[3 marks]

(e) The Haber–Bosch process generates 2% of the world's CO_2 emissions and consumes 1% of the world's energy. Indicate TWO ways in which CO_2 is emitted as a by-product of the Haber–Bosch process.

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[2 marks]

(f) State the effects of the MAJOR limitations if the reaction in the Haber–Bosch process is done at STP?

Effect 1 .....

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Effect 2 .....

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[2 marks]

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(g) Identify TWO ways in which ammonia affects the environment.

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[2 marks]

(h) As a fertilizer, ammonia is generally used in the form of an ammonium salt.

(i) State the reason for this.

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(ii) Give the formula for any ONE of the most commonly used ammonium salts in fertilizers.

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[2 marks]

Total 30 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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